

CUSTOMER ANALYTICS REPORT

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BUSINESS PROBLEM

Retail businesses collect large volumes of customer transaction data; however, without proper analysis, valuable business opportunities often remain hidden. Management needs to understand:

- Which customer segments generate the most revenue?
- Which product categories perform best?
- How effective is the subscription program?
- Which geographic locations contribute the highest sales?
- How satisfied are customers with their shopping experience?

Answering these questions enables better inventory planning, marketing decisions, and customer retention strategies.



PROJECT OVERVIEW

The primary objectives of this project are to:

- Analyze customer demographics and purchasing behavior.
- Evaluate revenue across different product categories.
- Identify high-value customer segments.
- Measure customer subscription trends.
- Examine regional revenue distribution.
- Monitor customer satisfaction using review ratings.
- Develop an interactive dashboard for business decision-making.
- Provide actionable recommendations based on data insights.

DATASET OVERVIEW

Attribute	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Male/Female
Category	Product category purchased
Purchase Amount	Total transaction value
Location	Customer state
Subscription Status	Membership status (Yes/No)
Review Rating	Customer satisfaction rating
Shipping Type	Delivery method selected
Season	Season in which purchase occurred

The dataset provides sufficient information to analyze purchasing behavior, customer segmentation, and revenue performance.

DATA ANALYTICS PIPELINE



PYTHON

- HANDLING MISSING VALUES IN REVIEW RATING
- CREATED AGE GROUP AND PURCHASE FREQUENCY COLUMNS



SQL

- REVENUE BY GENDER
- HIGH SPENDING DISCOUNT USERS
- TOP 5 PRODUCT BY RATING
- SHIPPING TYPE COMPARISON
- SUBSCRIBERS V/S NON-SUBSCRIBERS
- DISCOUNT DEPENDENT PRODUCTS
- CUSTOMER SEGMENTATION
- TOP 3 PRODUCTS PER CATEGORY
- REPEAT BUYERS AND SUBSCRIPTIONS
- REVENUE BY AGE GROUP



POWER BI

BUILT AN INTERACTIVE DASHBOARD IN POWER BI TO PRESENT INSIGHTS VISUALLY USING CHARTS AND SLICERS.

KEY FINDINGS

- Female customers generate slightly higher total revenue than male customers

- Express shipping customers spend 12% more per transaction

- Blouses and dresses achieved the highest customer ratings (5/5), reflecting exceptional customer satisfaction

- Subscribers consistently spend more, contribute significantly to revenue, and exhibit stronger purchase loyalty.

- Half of the customers are first-time buyers, highlighting a strong opportunity to convert new customers into loyal, high-value customers.

- Clothing generates the highest revenue, followed by Accessories, making these the business's strongest-performing product categories.

- Middle-aged customers generate the highest revenue and shop most frequently, making them the business's most valuable customer segment.

- Montana generates the highest revenue, followed by Illinois, California, Idaho, and Nevada, making them the business's top-performing markets.

BUSINESS RECOMMENDATIONS

Expand High-Performing Categories

- Focus on Clothing & Accessories
- Cross-selling and bundle offers

Strengthen Customer Retention

- Convert New → Returning → Loyal customers
- Reward loyal customers

Target High-Value Customers

- Prioritize middle-aged customers
- Personalized marketing

Increase Subscription Adoption

Loyalty rewards & exclusive benefits

Optimize Regional Strategy

Focus on Montana, Illinois, California, Idaho & Nevada

Enhance Customer Experience

- Promote top-rated products
- Improve lower-rated products

CONCLUSION

- ✓ Clothing & Accessories drive the highest revenue
- ✓ Middle-aged customers are the most valuable segment
- ✓ Top-performing states:
Montana • Illinois • California • Idaho • Nevada
- ✓ Increase subscriptions and customer retention
- ✓ Data-driven decisions support sustainable growth

THANK YOU